

NIRBHAY KATIYAR

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Backend Developer and Security Engineer with experience building scalable, distributed systems and conducting offensive security assessments. Skilled in REST API design, cloud-native AWS architecture, and web application security (OWASP Top 10). Proven track record of disclosing critical vulnerabilities at Spotify, IBM, and LightSpark via HackerOne.

TECHNICAL SKILLS

Languages & Frameworks: Python, C++, FastAPI, Flask, SQLAlchemy, PyDantic, REST APIs
Cloud & DevOps: AWS (Lambda, EventBridge, S3, DynamoDB), Docker, Serverless, Microservices, Linux
Databases: PostgreSQL, Redis, MongoDB, DynamoDB
Security: OWASP Top 10, Penetration Testing, JWT Hardening, SQLi, XSS, CSRF, SSRF, CORS, Burp Suite, NMAP
CS Fundamentals: Operating Systems, Computer Networking, DBMS, OSI/TCP-IP, System Design, OOP, LLD

EXPERIENCE

Offensive Cybersecurity Intern

May 2025 – Aug 2025

InlighnX Tech

Remote

- Executed **black-box and grey-box penetration tests** on production web applications, uncovering **15+ high/critical severity vulnerabilities** including SSRF, broken authentication, and privilege escalation flaws.
- Authored **security assessment reports** with reproducible PoCs, CVSS scores, and prioritized remediation steps, streamlining developer handoff and reducing mean time to patch.
- Applied **OWASP Top 10** methodology across REST API and authentication assessments, covering injection flaws, broken access control, and security misconfigurations.

Independent Security Researcher (Bug Bounty)

Feb 2024 – Jan 2025

HackerOne

Remote

- Discovered and responsibly disclosed **critical vulnerabilities** in platforms operated by **Spotify, LightSpark, and IBM**, earning valid accepted reports across all three organizations.
- Researched complex attack chains involving **IDOR, authentication bypass, SSRF, and information disclosure**, demonstrating real-world exploitability with clear business impact.

PROJECTS

Scalable URL Shortener | Python, PostgreSQL, Redis, Base62, CDN, Security Protocols | link.zerodaily.in

- Engineered a live URL shortener with **Base62 ID generation**; achieved **10ms server-side processing time** (25ms total with FastAPI overhead) and **600ms–900ms round-trip latency**.
- Implemented a **dual-layer storage architecture** utilizing PostgreSQL for persistent transactional storage and Redis for high-speed caching, drastically cutting query latency.
- Hardened deployment with **SSRF protection** for target resolution and global CDN integration (Cloudflare) for DDoS resilience and edge caching.

ZeroDaily | Python, Flask, AWS Lambda, EventBridge, S3, DynamoDB, OpenAI | zerodaily.in

- Architected a **fully serverless, event-driven news aggregator** that fetches, summarizes, and delivers curated security and tech newsletters using **LLM-powered summarization** — serving **100+ active subscribers**.
- Designed an **automated delivery pipeline** using AWS EventBridge and Lambda, achieving reliable email dispatch with **sub-minute execution latency** at near-zero infrastructure cost.

TOKELY - A Hybrid Stateless Auth Library | Python, Argon2, JWT | pypi.org/project/tokenly-auth/

- Built a **hybrid-state authentication library** in Python supporting **database adapters and JWT** implementations, enabling seamless custom middleware via built-in decorators.
- Integrated **robust hashing and encryption** standards using Argon2 for passwords and HTTP-only cookies to securely store JWTs with refresh token support and a token blacklisting mechanism.

EDUCATION

Galgotias College of Engineering & Technology

B.Tech in Electronics & Communication Engineering (Expected Apr 2029)

Greater Noida, UP

CGPA: 7.59